Kinematics 1D and 2D Revision Sheet

Level 3 Sheet **Maximum Time Taken to Solve: 40 Min**

Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 4.

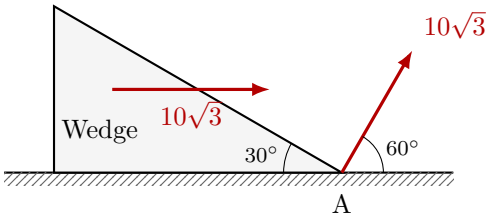
1. A ship is moving westward with a speed of 10 km/h and a ship B, 100 km south of A is moving northwards with same speed. The time after which the distance between them is shortest and the shortest distance are:

(1) 2h, 100 km (2) 5h, $50\sqrt{2}$ km (3) $5\sqrt{2}$ h, 50 km (4) $10\sqrt{2}$ h, $50\sqrt{2}$ km

2. A man can swim in still water with a velocity 5 m/s. He wants to reach at directly opposite point on the other bank of a river which is flowing at a rate of 4 m/s. River is 15 m wide and the man can run with twice the velocity as compared with velocity of swimmer with respect to river. If he swims perpendicular to river flow and then run along the bank, then time taken by him to reach the opposite point is :

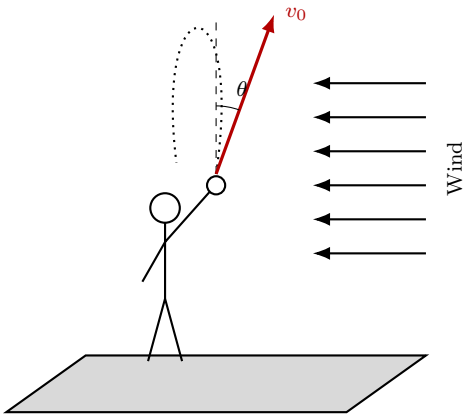
(1) 3 sec (2) less than 3 sec (3) 5 sec (4) 4.2 sec

3. A particle is projected at angle 60° with speed $10\sqrt{3}$, from the point 'A' as shown in the fig. At the same time the wedge is made to move with speed $10\sqrt{3}$ towards right as shown in the figure. Then the time after which particle will strike with wedge is ($g = 10$ m/sec^2) :



(1) 2 sec (2) $2\sqrt{3}$ sec (3) $\frac{4}{\sqrt{3}}$ sec (4) none of these

4. A boy throws a ball upwards with velocity $v_0 = 20$ ms^{-1}. The wind imparts a horizontal acceleration of 4 ms^{-2} to the left. The angle θ at which the ball must be thrown so that the ball returns to the boy's hand is ($g = 10$ ms^{-2})



(a) $\tan^{-1}(1.2)$ (b) $\tan^{-1}(0.2)$

(c) $\cot^{-1}(2)$ (d) $\cot^{-1}(2.5)$

5. A river is flowing from west to east direction with speed of 9 km h^{-1}. If a boat capable of moving at a maximum speed of 27 km h^{-1} in still water, crosses the river in half a minute, while moving with maximum speed at an angle of 150° to direction of river flow, then the width of the river is :

(a) 300 m (b) 112.5 m

(c) 75 m (d) $112.5 \times \sqrt{3}$ m

6. A person is swimming with a speed of 10 ms^{-1} at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is x ms^{-1}. The value of x to the nearest integer is _____.

7. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}$ km h^{-1}, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is

(a) 30 km h^{-1} (b) $\frac{25}{\sqrt{2}}$ km h^{-1}

(c) $\frac{30}{\sqrt{2}}$ km h^{-1} (d) 25 km h^{-1}

Keep Going!

"The harder you work for something, the greater you will feel when you achieve it. Stay consistent — your effort is adding up."